Supplemental Material for "Exact exponential heat rectification from the Liouvillian skin effect"

Hikaru Wakaura[1] and Taiki Tanimae[1]

[1] *QIRI (Quantum Integrated Research Institute Inc.), Tokyo 107-0061, Japan*

S1. GAUSSIAN SOLUTION AND THE ΔT RESPONSE

The Lindbladian of the main text, $\dot\rho = -i[H,\rho] + \sum_a \mathcal{D}[L_a]\rho$, is quadratic for $V = 0$: $H = \sum_{mn} h_{mn} c_m^\dagger c_n$ and every jump operator is linear, $L_a = \sum_m l_{a,m} c_m$ (loss) or $L_a = \sum_m g_{a,m} c_m^\dagger$ (gain). The leads are included explicitly as sets of modes with their own loss and gain jumps whose rates realise a thermal occupation. The two-point function $C_{mn} = \langle c_m^\dagger c_n \rangle$ then obeys a closed equation,

$$\dot C = -(XC + CX^\dagger) + M_g^T, \qquad X = \tfrac{1}{2}(M_l + M_g^T) - ih^T, \qquad M_l = \sum_a l_a l_a^\dagger, \quad M_g = \sum_a g_a g_a^\dagger, \tag{S1}$$

whose steady state is the solution of the Lyapunov equation $XC + CX^\dagger = M_g^T$. We solve it with a Bartels–Stewart solver (`scipy.linalg.solve_continuous_lyapunov`) and verified it against the exact steady state of the full many-body Lindbladian on four modes to 10^{-8}. The energy current from reservoir α into the system is obtained from the jump operators labelled α and from h and C; the single-particle amplitudes propagate with $h_{\rm eff} = h - \frac{i}{2}(M_l^T + M_g)$, which for pure bond loss is Eq. (2) of the main text.

Leads. Two implementations are used. (i) *Wide-band leads*: the end sites acquire a self-energy $-i\Gamma/2$ and the heat currents are given by the Landauer formula with $T_{RL}(\varepsilon) = \Gamma^2 |G_{L1}(\varepsilon)|^2$, $T_{LR}(\varepsilon) = \Gamma^2 |G_{1L}(\varepsilon)|^2$, $G = (\varepsilon - h_{\rm eff})^{-1}$; we use $\Gamma = J$ and integrate over $|\varepsilon| \le 4J$. (ii) *Mesoscopic leads*: K modes equally spaced in $[-W, W]$, each coupled to its end site with $v = \sqrt{\Gamma \Delta\varepsilon / 2\pi}$ and thermalised by loss and gain rates $\gamma(1-f)$ and γf, with f the Fermi function at (T_α, μ) and $\gamma = \Delta\varepsilon$ chosen to make the lead spectrum smooth. We use $K = 10$, $W = 4J$ for the matrix-product-state (MPS) runs and $K = 10$–80 for checks. The ΔT responses of the two implementations agree within 5–15%, the difference being a bandwidth effect; the ratio $\delta Q_f / \delta Q_r$ obeys Eq. (3) exactly in both, since only the end self-energies differ and they are diagonal.

Why total currents are the wrong observable. With bulk loss, the total heat current absorbed by the cold lead contains a contribution that does not depend on the temperature of the other lead: both leads lose energy to the loss channel. If rectification is defined from total currents, this

background dominates and the apparent $\mathcal{R}$ decays as $1/L$ for any κ, although the device is perfectly directional. The ΔT responses $\delta Q_f = \dot{E}_R(T_h, T_c) - \dot{E}_R(T_c, T_c)$ and $\delta Q_r = \dot{E}_L(T_c, T_h) - \dot{E}_L(T_c, T_c)$ (heat absorbed by the respective cold lead) remove the background exactly and are the quantities whose ratio obeys the exact law. Throughout, $T_h = J$, $T_c = 0.5J$ and $\mu = 0$ unless stated otherwise; $\hbar = k_B = 1$.

S2. PROOFS OF THE EXACT STATEMENTS

Tridiagonal identity. For a tridiagonal matrix A of dimension L with sub-diagonal $a_{j+1,j}$ and super-diagonal $a_{j,j+1}$, the corner elements of the inverse are $(A^{-1})_{L1} = (-1)^{L+1} \prod_j a_{j+1,j} / \det A$ and $(A^{-1})_{1L} = (-1)^{L+1} \prod_j a_{j,j+1} / \det A$ (expand the cofactor of the corner element; the only non-vanishing term is the product of the off-diagonal chain). With $A = \varepsilon - h_{\text{eff}}$ this gives $G_{L1}/G_{1L} = \prod_j (h_{\text{eff}})_{j+1,j} / (h_{\text{eff}})_{j,j+1}$. The product involves no diagonal element, so it is independent of ε, of the on-site energies, of the end self-energies and of the uniform loss. For the uniform chain all factors equal t_-/t_+ and $T_{RL}/T_{LR} = |t_-/t_+|^{2(L-1)}$ at every energy. The thermal averages $\delta Q_{f,r} = \int \frac{d\varepsilon}{2\pi} \varepsilon T_{RL,LR}(\varepsilon) \Delta f(\varepsilon)$ share the same weight, hence Eq. (3). The argument does not use the fermionic nature of the leads: for bosonic carriers with the same bond loss Δf is a difference of Bose functions and the ratio is unchanged.

Bond disorder and the sum rule. With bond-dependent κ_j and ϕ_j the factors are $t_{\mp,j} = -J - \frac{i\kappa_j}{2} e^{\mp i\phi_j}$ and the identity gives Eq. (5) of the main text directly. We verified the sum rule to 10^{-14} on 10^2 random samples, with on-site disorder present at the same time.

Same-chirality gain. Adding $G_j = \sqrt{\kappa g}\,(c_j^\dagger + e^{-i\phi} c_{j+1}^\dagger)$ produces M_g with the same matrix elements as M_l^T, so h_{eff} is obtained from the pure-loss one by $\kappa \to \kappa(1+g)$: the law keeps its form with a shorter ξ. Consequently a local bath at infinite temperature and $\mu_b = 0$, implemented as loss $\kappa/2$ plus gain $\kappa/2$ of the same chirality, has exactly the free-fermion ratio of a pure loss κ. This is the configuration used for the green symbols in Fig. 3 of the main text.

Mirror reflection equals transposition and the identity $\sigma_f(T_b = T_c) = \sigma_f(T_b = T_h)$. Let P be the site reflection $j \to L + 1 - j$. For the uniform chiral chain $P h_{\text{eff}} P$ exchanges t_+ and t_-, which is also what transposition does, so $P h_{\text{eff}} P = h_{\text{eff}}^T$ and $PGP = G^T$. The end reflectances therefore coincide, $G_{LL} = (PGP)_{11} = (G^T)_{11} = G_{11}$, although $G_{L1} \neq G_{1L}$. In the multi-terminal Landauer form with all reservoirs thermal, the entropy production with every reservoir except L at T_c is $\sigma = (1/T_c - 1/T_h) \int \frac{d\varepsilon}{2\pi} \varepsilon\,[f_h - f_c]\,[1 - R_{LL}(\varepsilon)]$, and with every reservoir except R at T_h it is the same expression with R_{RR} (the sum rules $\sum_\beta T_{\alpha\beta} = \sum_\beta T_{\beta\alpha} = 1 - R_{\alpha\alpha}$ are used). Equal

reflectances therefore give $\sigma_f(T_b = T_c) = \sigma_f(T_b = T_h)$ for any ϕ, κ and t_2. On-site disorder breaks the identity (difference 3×10^{-3} for $W_d = J$), and mirror-symmetric disorder restores it (10^{-12}); the clean difference is 10^{-12}. Physically, the heat emitted by a hot lead into an otherwise cold device equals the heat absorbed by a cold lead from an otherwise hot device: nonreciprocity appears in transmission but not in reflection.

S3. NON-BLOCH FORM OF THE EXPONENT FOR NEXT-NEAREST-NEIGHBOUR HOPPING

With $t_2 \neq 0$ the bulk ansatz $\psi_j = \beta^j$ turns $h_{\text{eff}}\psi = \varepsilon\psi$ into the quartic Eq. (4) of the main text. Its roots satisfy $\beta_1\beta_2\beta_3\beta_4 = 1$ by Vieta's formula. For a long open chain the transfer matrix from site 1 to site L is dominated by the two roots of intermediate modulus, which form the generalised Brillouin zone when $|\beta_2| = |\beta_3|$ and otherwise split it; the amplitude transported from left to right scales as $|\beta_3|^L$ and from right to left as $|\beta_2|^{-L}$, giving $\ln[T_{RL}/T_{LR}](\varepsilon) = 2(L-1)\ln|\beta_2\beta_3|(\varepsilon)+O(1)$. For $t_2 = 0$ the equation is quadratic with $\beta_1\beta_2 = t_-/t_+$, and the $O(1)$ term vanishes identically. Figure 2(a) of the main text compares the prediction with the exact transmissions at $\kappa = 0.2J$; the agreement is equally good at $\kappa = 0.1J$ and $0.3J$ for all $t_2 \leq 0.5J$.

The periodic-boundary spectrum is $E(k) = -2J\cos k + 2t_2\cos 2k - i\kappa(1-\sin k)$ for $\phi = \pi/2$. Its imaginary part is non-positive and vanishes only at $k = \pi/2$, so the loop touches the real axis at one point and the winding number $w(\varepsilon) = \frac{1}{2\pi i}\oint d\ln[E(k)-\varepsilon]$ is zero for every real ε and every t_2. The winding around the centre of the loop, $-i\kappa$, is -1 and signals the skin effect, but it is a single integer and cannot encode the energy dependence, sign change and t_2 dependence of the transmission asymmetry; the quantity that does is $\ln|\beta_2\beta_3|(\varepsilon)$.

S4. SUPERFERMION MATRIX-PRODUCT-STATE METHOD AND ITS PRECISION

Vectorisation. We use the superfermion representation: $|\rho\rangle\rangle = \rho|I\rangle\rangle$ with $|I\rangle\rangle = \prod_j(1+c_j^\dagger\tilde{c}_j^\dagger)|0\rangle$, where $\tilde{c}_j$ are auxiliary (bra) fermions. For a parity-even ρ the rules $c_j\rho \to c_j$, $\rho c_j^\dagger \to -\tilde{c}_j$, $\rho c_j \to \tilde{c}_j^\dagger$, $\rho c_i^\dagger c_j \to \tilde{c}_j^\dagger\tilde{c}_i$ map the Lindbladian onto a local operator on $2N$ fermionic modes; the bond dissipators $L_j\rho L_j^\dagger$ become bilinears on four modes and the interaction Vn_jn_{j+1} appears as $Vn_jn_{j+1} - V\tilde{n}_j\tilde{n}_{j+1}$. Each physical site is grouped with its auxiliary partner into one MPS site of local dimension four, and the conserved charge $N - \tilde{N} = 0$ is exploited. Expectation values are $\langle\langle I|O|\rho\rangle\rangle/\langle\langle I|\rho\rangle\rangle$. The lead jump operators commute with the interaction, so the lead energy

currents are two-point functions and the Gaussian current formula applies unchanged.

Layout and propagation. The L chain sites and the $2K$ lead modes are arranged as [site 1][left lead][sites $2 \dots L$][right lead], which keeps all long-range couplings anchored at a chain end and the MPO bond dimension of the Lindbladian at six. The state is propagated with the second-order W^{II} MPO exponential of TeNPy in steps dt, with SVD compression at bond dimension χ and singular-value cutoff 10^{-12}, starting from the product state with the exact Gaussian occupations, until the change of both lead currents over a chunk of $2/J$ is below 10^{-5} or $t = 60/J$ is reached. The mapping was verified exactly: on four modes with $V = 0.7J$ the null vector of the MPO Lindbladian coincides with the many-body steady state to 10^{-15}, and for $V = 0$, $L = 6$, $K = 4$ the propagated steady state agrees with the Lyapunov solution to 2×10^{-3} in the currents.

Precision. The reverse response δQ_r is exponentially small and is the limiting quantity. The truncation error of the absolute currents accumulates with the number of time steps, so it is *not* reduced by a smaller dt; it is reduced only by χ. Table S1 lists the free-fermion calibration at $\kappa = 0.05J$, $L = 16$. At $\chi = 256$ and $dt = 0.05/J$ the ratio is reproduced within 0.003 and δQ_r within 0.03%; this setting is used for all production runs. At $\kappa = 0.1J$ the residual bias of $\ln(\delta Q_f/\delta Q_r)$ at $\chi = 256$ is $+0.045$ ($L = 12$) and -0.038 ($L = 16$), which we quote as the error bar; slopes at this κ were extracted from the control-variate estimate $\ln r_{\mathrm{cv}}(V) = \ln r_{\mathrm{MPS}}(V) - \ln r_{\mathrm{MPS}}(0) + \ln r_{\mathrm{exact}}(0)$, which cancels the common truncation bias. The time-fluctuation of the last five chunks contributes less than 0.01 to $\ln(\delta Q_f/\delta Q_r)$ except for the infinite-temperature-bath run at $V = 2J$, where it is 0.05.

TABLE S1. Free-fermion calibration of the MPS steady state at $\kappa = 0.05J$, $L = 16$, $K = 10$: logarithmic ratio and reverse response compared with the exact Gaussian values $\ln(\delta Q_f/\delta Q_r) = 1.5003$ and $\delta Q_r = 4.918 \times 10^{-3} J^2/\hbar$.

	$\chi = 128$, $dt = 0.1$	$\chi = 128$, $dt = 0.05$	$\chi = 128$, $dt = 0.025$	$\chi = 256$, $dt = 0.05$
$\ln(\delta Q_f/\delta Q_r)$	1.347	1.412	1.462	1.5033
δQ_r bias	+16.8%	+7.7%	+2.6%	+0.03%

Resolvable regime. Because the MPS floor of the absolute currents is $\sim 10^{-3} J^2/\hbar$ at $\chi = 128$ and about two orders of magnitude lower at $\chi = 256$, the reverse response can be resolved only when $\delta Q_r/\delta Q_f = e^{-2(L-1)/\xi}$ is not much smaller than 10^{-2}, i.e. $2(L-1)\kappa/J \lesssim 4$. For stronger skin effect (e.g. $\kappa = 0.2J$, $L = 20$–40) the simulations yield only lower bounds, $\mathcal{R} \geq 0.85$–0.98 for all $V \leq 2J$, together with the forward throughput, which decreases monotonically with V ($1.10 \to 0.63 \times 10^{-2} J^2/\hbar$ at $L = 20$ and $0.70 \to 0.33 \times 10^{-2}$ at $L = 40$ for $V = 0 \to 2J$).

Interacting results. Table S2 collects the $\chi = 256$ values used in Fig. 3. The $\chi = 128$ series at $\kappa = 0.03J$ and $0.05J$ over $L = 12, 16, 20$ give slopes $2/\xi_{\rm eff}$ of $0.055, 0.066, 0.046, 0.043 \pm 0.013$ ($\kappa = 0.03J$; exact 0.060) and $0.096, 0.101, 0.125, 0.094 \pm 0.019$ ($\kappa = 0.05J$; exact 0.100) for $V = 0, 0.5, 1, 2J$. A direct linear-response evaluation of $\delta Q_{f,r}$ by integrating the perturbed Lindbladian dynamics was also implemented; it is exact for $N \leq 14$ modes but, at equal χ, less accurate than the subtraction method for $N \geq 32$, and was not used.

TABLE S2. $\chi = 256$ steady-state results ($dt = 0.05/J$, $K = 10$, $\phi = \pi/2$). "bath" denotes the local infinite-temperature bath (loss $\kappa/2$ + gain $\kappa/2$, total rate $0.1J$).

κ/J	L	exact $V = 0$	MPS $V = 0$	MPS $V = 2J$	Δ_V	slope $V = 2J$ / exact
0.05	16	1.5003	1.5033	1.083	−0.42	0.0975/0.1000
0.05	20	1.9004	—	1.473	−0.43	
0.1	12	2.2018	2.247	1.704	−0.54	0.207/0.200
0.1	16	3.0025	2.964	2.448	−0.52	
0.1 (bath)	12	2.2018	2.203	1.662	−0.54	—

S5. FINITE-TEMPERATURE BULK BATH

Each bond j is coupled to K_b bath modes with energies in $[-W_b, W_b]$, thermalised at (T_b, μ_b) as in the leads, through the bond operator $A_j = c_j + e^{i\phi} c_{j+1}$ with amplitude $v_b = \sqrt{\kappa \Delta\varepsilon_b / 2\pi}$, so that the damping rate of A_j is κ and the pure-loss model is recovered for $\mu_b \to -\infty$. All reservoirs now have a temperature and $\sigma = -\sum_\alpha (\dot{E}_\alpha - \mu_\alpha \dot{N}_\alpha)/T_\alpha$ is defined; we verified $\sigma \geq 0$ in all configurations and $\sigma = 0$ to 10^{-8} when all reservoirs share one temperature. We use $L = 40$, $K_{\rm lead} = 10$, $K_b = 10$, $W_b = 4J$.

Because the bath self-energy is energy dependent, the effective $t_\pm(\varepsilon)$ are not constant and the ratio falls slightly below the wide-band law: at $\kappa = 0.2J$ we obtain $\ln(\delta Q_f / \delta Q_r) = 14.23$, 14.48, 14.68 and 14.95 for $(K_b, W_b/J) = (10, 4)$, $(10, 6)$, $(20, 4)$ or $(20, 6)$, and $(40, 6)$, approaching the wide-band value 15.65. The Lindblad loss is the exact wide-band limit. The ratio is identical for all (T_b, μ_b) at fixed discretisation (difference $< 10^{-9}$).

With $T_b = T_c$ and $\mu_b = 0$ the isothermal entropy production vanishes and $\sigma_f = 0.042$, 0.044, 0.047, 0.049, $0.052\, J/\hbar$ for $\kappa = 0.05$, 0.1, 0.2, 0.3, $0.5J$, while $\delta Q_f = 1.64$, 1.11, 0.66, 0.47, $0.29 \times 10^{-2} J^2/\hbar$; the cost $\sigma_f / \delta Q_f = 2.6$, 4.0, 7.0, 10.4, $18.1\, J^{-1}$ is the quantity in Fig. 4(a). The cost is carried by the heat flowing from the hot lead into the bath, $\dot{Q}_{{\rm bulk},f} = 0.026 \to 0.049\, J^2/\hbar$

over the same range. For $\kappa = 0.2J$ the μ_b sweep gives $\sigma_f = 0.047$, 0.12, 0.34, 1.15, 3.2, 25.5 $J/\hbar$ and mean filling 0.50, 0.44, 0.38, 0.25, 0.12, 0.10 for $\mu_b = 0, -0.5, -1, -2, -4, -30J$; the skin-like density pile-up at the edge appears only for $\mu_b < 0$. For $T_b \neq T_c$ the isothermal state already produces entropy through bath–lead heat flow and $\sigma_f - \sigma_{\rm iso}$ can be negative; the cost indicator is therefore quoted at $T_b = T_c$.

S6. ADDITIONAL DISORDER DATA

On-site disorder $\varepsilon_j \in [-W_d/2, W_d/2]$ (uniform), $L = 40$, wide-band leads, 60–100 samples per point. The rectification is sample independent to 10^{-14}: $\mathcal{R} = 0.9604$, 0.9992, 1.0000 for $\kappa = 0.05$, 0.1, $0.2J$ at every W_d. The median forward throughput at $\kappa = 0.1J$ is 1.62, 1.26, 0.61, 0.051, 0.0014, $0.00016 \times 10^{-2} J^2/\hbar$ for $W_d = 0$, 0.5, 1, 2, 3, $4J$. The forward decay rate $-d\langle \ln \delta Q_f \rangle / dL$ between $L = 10$ and 80 is listed in Table S3; in the weak-disorder regime the loss-induced and Anderson decay rates are nearly additive, and no "delocalisation by the skin effect" appears in the forward direction, which is attenuated by loss and localisation alike. Bond disorder at $\kappa = 0.1J$: a spread $\kappa_j = \kappa(1 \pm 0.8u)$, u uniform in $[0, 1]$, gives means 3.75, 7.85, 15.83 (clean 3.80, 7.81, 15.81) and standard deviations 0.43, 0.55, 0.88 for $L = 20$, 40, 80; a phase spread $\phi_j = \pi/2 \pm 1$ gives means rescaled by $\langle \sin \phi_j \rangle = \sin 1 \simeq 0.84$ (6.56 against $7.81 \times 0.84 = 6.57$ for $L = 40$) and standard deviations 0.13–0.25; random chirality $\phi_j = \pm\pi/2$ gives means -0.07, -0.04, 0.16 and standard deviations 0.91, 1.34, 1.77. With $t_2 = 0.3J$ and $\kappa = 0.2J$ the ratio becomes sample dependent, $\ln(\delta Q_f / \delta Q_r) = 10.12 \pm 0.11$, 9.71 ± 0.19, 9.42 ± 0.30 for $W_d = 0.5, 1, 2J$ (clean 10.31).

TABLE S3. Forward decay rate $-d\langle \ln \delta Q_f \rangle / dL$ (per site, $L = 10 \to 80$) for on-site disorder of width W_d.

	$W_d = 0.5J$	$W_d = J$	$W_d = 2J$
$\kappa = 0$	0.006	0.020	0.055
$\kappa = 0.05J$	0.017	0.034	0.090
$\kappa = 0.2J$	0.027	0.044	0.107

S7. EXPERIMENTAL PARAMETERS AND RELATED WORK

Table S4 summarises the parameter sets quoted in the main text. The requirement $\mathcal{R} > 0.99$ corresponds to $(L-1)/\xi > 2.65$ with $\xi \simeq J/\kappa$ at $\phi = \pi/2$. Related but distinct results are the diffusive skin effect in classical thermal metamaterials [1], which concerns heat conduction

rather than quantum transport between reservoirs; non-reciprocal routing of thermal noise in few-mode cascaded networks [2], which does not scale with system size; the transport theory of non-Hermitian nonreciprocal systems [3], which treats reflectionless and transparent transport but not the thermal rectification ratio; and the recent photonic observation of skin dynamics in the Liouvillian regime [4]. For a review of boundary-driven open quantum systems see Ref. [5].

TABLE S4. Parameter sets for the two proposed platforms.

	cold atoms (momentum lattice)	superconducting circuit
$J/2\pi$	0.5 kHz	10–50 MHz
auxiliary damping $\Gamma_a/2\pi$	optical loss	~ 10 MHz
coupling $g/2\pi$	Raman	1–3 MHz
$\kappa = 4g^2/\Gamma_a$, $\kappa/2\pi$	25–100 Hz	0.5–2 MHz
κ/J	0.05–0.2	0.05–0.2
L for $\mathcal{R} > 0.99$	15 ($\kappa/J = 0.2$) to 60 (0.05)	15 to 60
thermal leads	two-terminal reservoirs [6, 7]	noise temperature 20–300 mK on end lines
carriers	fermionic atoms	photons (law unchanged, Sec. S2)

[1] P.-C. Cao, Y. Li, Y.-G. Peng, M. Qi, W.-X. Huang, P.-Q. Li, and X.-F. Zhu, Commun. Phys. **4**, 230 (2021).

[2] A. Xuereb, M. Aquilina, and S. Barzanjeh, in *Proc. SPIE*, Vol. 10672 (2018) p. 10672N.

[3] H. Ghaemi-Dizicheh and H. Schomerus, Phys. Rev. A **104**, 023515 (2021).

[4] S. Yang, Y. Sun, L. Hong, and Y. Yang, arXiv:2606.27043 (2026).

[5] G. T. Landi, D. Poletti, and G. Schaller, Rev. Mod. Phys. **94**, 045006 (2022).

[6] J.-P. Brantut, J. Meineke, D. Stadler, S. Krinner, and T. Esslinger, Science **337**, 1069 (2012).

[7] S. Krinner, T. Esslinger, and J.-P. Brantut, J. Phys.: Condens. Matter **29**, 343003 (2017).